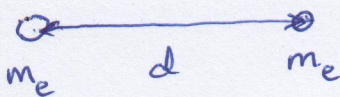


HW2

(1)

$$Q = -e$$

$$Q = -e$$



$$F_G = G \frac{m^2}{d^2}$$

$$F_E = k_e \frac{e^2}{d^2}$$

$$R = \frac{F_G}{F_E} = \frac{G m_e^2}{k_e e^2}$$

In SI units,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

$$k_e = 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

$$e = 1.60 \times 10^{-19} \text{ C}$$

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

$$\text{So } R = 2.41 \times 10^{-43}$$

gravity is extremely weak compared to electromagnetism.

(2)

$$x = vt$$
$$= (\beta c)t$$

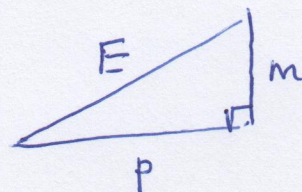
$$t = \frac{x}{\beta c}$$

So need to calculate β .
implicit in

$$E^2 = p^2 + m^2$$

Obtain this from "the Lorentz" triangle

(natural units).



$$\frac{E^2}{m^2} = \frac{p^2}{m^2} + 1$$

$$\frac{E^2}{E^2} = \frac{p^2}{E^2} + \frac{m^2}{E^2}$$

$$1 = \beta^2 + \frac{1}{\gamma^2}$$

Remember, the definition of

$$\frac{E}{m} = \gamma$$

$$\frac{p}{E} = \beta$$

$$\frac{p}{m} = \gamma\beta$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

$$\text{So } \gamma^2 = \frac{1}{(1-\beta^2)}$$

$$\gamma^2 - \gamma^2\beta^2 = 1$$

$$\gamma^2\beta^2 = \gamma^2 - 1$$

$$\beta^2 = \frac{\gamma^2 - 1}{\gamma^2} = 1 - \frac{1}{\gamma^2} = 1 - \left(\frac{m}{E}\right)^2 = \frac{E^2 - m^2}{E^2} = \frac{p^2}{E^2}$$

$$\text{So } \boxed{\beta = \frac{p}{E}}$$

<u>Particle</u>	<u>Mass (GeV)</u>	<u>Energy* (GeV)</u>	<u>$\beta (\frac{p}{E})$</u>	<u>Time (ns)</u>
e^-	0.511×10^{-3}	1.000080131	0.999999869	
μ^-	0.1057	1.00557	0.99446	
π^-	0.1396	1.009697	0.990396	
K^-	0.4937	1.1152	0.896675	
p	0.9383	1.3713	0.72925	

* for $p = 1 \text{ GeV}/c$ using $E^2 = (p^2 + m^2)^{1/2}$

Using $c = 29.9792458 \text{ m/ns}$

$\Rightarrow$ Time-of-flight (ns)

$$(T = \frac{x}{\beta c}).$$

e	3.35 3.335	3.3355
μ	3.34	3.3541
π		3.3679
K		3.7199
p		4.5740

$\gamma (\beta=1)$

Typical detector timing resolution
 $\mathcal{O}(100 \text{ ps}) \rightarrow$ hard to use
time-of-flight over short
path lengths to identify all but the
lowest momentum particles.

③ Similarly.

for $p = 2 \text{ GeV}$

β values range

from $\beta_e \approx 1$ to $\beta_p = 0.90532$

$$c_{\text{medium}} = \frac{c}{n}$$

So $\checkmark$ radiation if $\beta > \frac{1}{n} = \frac{1}{1.33} = 0.752$.

So all 5 mass hypotheses do emit $\checkmark$ radiation,

with an angle given by $\theta = \cos^{-1}\left(\frac{1}{\beta n}\right)$

So for $p = 2 \text{ GeV}$

<u>Particle</u>	<u>$\theta_{\text{Cerenkov}} (\text{rad})$</u>	<u>(β)</u>
e	0.7199	0.99999997
μ	0.7183	0.99861
π	0.7171	0.99757
K	0.6850	0.9709
p	0.5908	0.9053.

β values of e, μ, π very similar, so not much difference in the Cerenkov angle.

④ $\check{c}$ momentum thresholds.

$$\cos \theta_c = \frac{1}{\beta n}$$

When $\beta n = 1$ i.e. $\beta = \frac{1}{n}$, $\cos \theta_c = 1 \Rightarrow \theta_c = 0$

So for $\beta < \frac{1}{n}$; no $\check{c}$ light.

$$\beta = \frac{1}{n} \quad \text{So} \quad \frac{p}{(p^2 + m^2)^{1/2}} = \frac{1}{n}$$

Square $\frac{p^2}{p^2 + m^2} = \frac{1}{n^2}$

$$p^2 n^2 = p^2 + m^2$$

$$p^2 (n^2 - 1) = m^2$$

$$p^2 = \frac{m^2}{(n^2 - 1)}$$

$$p = \frac{m}{\sqrt{n^2 - 1}} = \frac{m}{\sqrt{(n+1)(n-1)}}$$

$x = n-1$ is small

$$= \frac{m}{\sqrt{(2+x)x}}$$

$$\approx \frac{m}{\sqrt{2x}} = \frac{m}{\sqrt{2(n-1)}}$$

So for e, μ, π, K, p , with $n-1 = 3.49 \times 10^{-5}$

$$P_{\text{thresh}} = \begin{matrix} 61.2 \text{ MeV} & 12.65 \text{ GeV} & 16.71 \text{ GeV} & 59.1 \text{ GeV} & 112.3 \text{ GeV} \\ (e) & (\mu) & (\pi) & (K) & (p) \end{matrix}$$

(5)

(a)

$$N(t) = N_0(t=0) e^{-t/\tau}$$

$$N(t=10\mu s) = N_0(t=0) e^{-10\mu s / 2.2\mu s.}$$

$$N_0 = 10^6$$

$$\Rightarrow N = 10^6 e^{-10/2.2} = 10,615 = 1.0615 \times 10^4$$

(b) Muons, $p = 45 \text{ GeV}$.

$$m = 0.1057 \text{ GeV}.$$

$$\gamma = \frac{E}{m} = \frac{(p^2 + m^2)^{1/2}}{m} = 425.7$$

Time dilation.

Reference frame of the problem is ~~a~~ one where the muon is moving, not where it is at rest.

Time intervals as measured between muon creation and decay are dilated, (lengthened) by the Lorentz γ factor.

$$\text{So } N(t) = N_0(t=0) e^{-t/\gamma\tau}$$

$$\Rightarrow N = 989,380$$

⑥

$$\gamma^5 \equiv i \gamma^0 \gamma^1 \gamma^2 \gamma^3 = \begin{pmatrix} 0 & I_{2 \times 2} \\ I_{2 \times 2} & 0 \end{pmatrix}$$

$$\gamma^2 \gamma^3 = i \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = i \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

$$\gamma^0 \gamma^1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & +1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

So ~~$\gamma^5 = (i \gamma^0 \gamma^1) (\gamma^2 \gamma^3)$~~

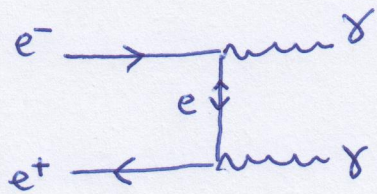
~~$= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$~~

$$\gamma^5 = \gamma^0 \gamma^1 (i \gamma^2 \gamma^3)$$

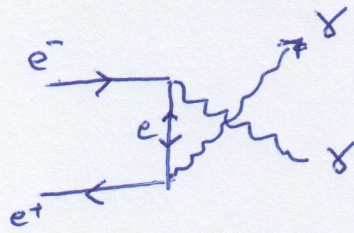
$$= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & +1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & I_{2 \times 2} \\ I_{2 \times 2} & 0 \end{pmatrix}$$

7



t-channel



u-channel
(identical particles in the final state).

8

$$\tau = \frac{1}{\Gamma} \quad \begin{array}{l} \uparrow \text{lifetime} \\ \uparrow \text{decay rate.} \end{array}$$

particle width, $\Gamma = \frac{\hbar}{\tau}$
(FWHM of Breit-Wigner resonance)
 $\uparrow$ MeV

$$\hbar = 6.582 \times 10^{-22} \text{ MeV s}$$

$$\tau_{\pi^0} = 8.4 \times 10^{-17} \text{ s.}$$

$$\begin{aligned} \Gamma_{\pi^0} &= \frac{\hbar}{\tau} = \frac{6.582 \times 10^{-22} \text{ MeV s}}{8.4 \times 10^{-17} \text{ s}} \\ &= 7.8 \text{ eV} \end{aligned}$$

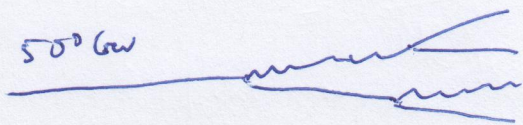
$$\tau_n = \frac{\hbar}{\Gamma} = \frac{6.582 \times 10^{-22} \text{ MeV s}}{1.3 \times 10^{-3} \text{ MeV}} = 5.1 \times 10^{-19} \text{ s}$$

⑨ Mean free path of photons is $\frac{9}{7} X_0$

Once electron ~~reach~~ the slow down to the critical energy, they lose most subsequent energy by ionization rather than radiation.

How many E_c electrons/positrons can we get from 500 GeV.

$$N_{\text{critical}} = \frac{500 \text{ GeV}}{7.97 \times 10^{-3} \text{ GeV}} = 62,735 \approx 2^{16} = 65,536.$$



X_0 is also the mean distance over which a high energy electron loses all but $\frac{1}{e}$ of its energy.

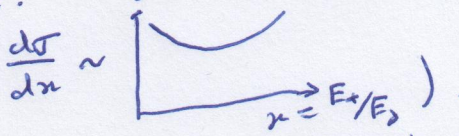
ie $E_{\text{remaining}} = E_0 e^{-x/X_0} = \frac{E_0}{e}$

$$E_{\text{lost (ie radiated)}} = E_0 - E_{\text{remaining}} = E_0 (1 - e^{-x/X_0})$$

$$= E_0 (1 - e^{-x/X_0})$$

(This also assumes symmetric energy partitioning ...)

ie $E_+ = E_- = \frac{1}{2} E_x$ but distribution is



$$= E_0 (1 - \frac{1}{e})$$

So taking the particle multiplicity doubling length as $\frac{9}{7} X_0$.

One requires 16 doublings or about $16 \times \frac{9}{7} X_0$ for all electrons (on average) to be at the critical energy, namely $21 X_0$.

adding on $1 X_0$ for ^{critical energy} electrons to stop, one gets an estimate of $22 X_0$.

There are fluctuations though. Sometimes the photon travels much further than $\frac{9}{7} X_0$ before converting. (An accurate answer needs simulation)

Anyway $\rightarrow$ ^{rough} estimate is ^{at least} $\approx 22 \times 0.35 \text{ cm} = 7.7 \text{ cm}.$

⑩ $\pi^0 \rightarrow \gamma\gamma$

$$BR(\pi^0 \rightarrow \gamma\gamma) = 0.98823 \pm 0.00034$$

Mean free path for high energy photons, $\lambda = \frac{9}{7} X_0$.

ie. $N(t = 0.5 \times_0) = N_0 e^{-t/T}$

↑
non-interacting
(ie non-converging).

$$\Rightarrow p_{\text{not condensing}}^{\text{per photon}} = \frac{N(t=0.5)}{N_0} = e^{-\frac{0.5}{9/7}}$$

Both photons not converting

$$\Rightarrow P = P_{\text{not-convert}}^2 = e^{-7/9}$$

$$\Rightarrow f_{\text{measurable}} = \frac{N_{\pi^0}(\text{produced}) \times BR(\pi^0 \rightarrow \gamma\gamma) \times P}{N_{\pi^0}(\text{produced})}$$

$$= 0.48823 \times e^{-7/9}$$

$$= 0.4540$$

11

BGO = scintillating crystal. ↑

Lead-Glass = block of heavy glass with

In both cases, an electromagnetic shower is initiated.

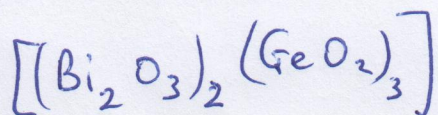
In BGO, the ionization leads to scintillation light. This is the primary source of visible photons.

In Lead-glass, there is no scintillation.

The detector response is mainly from relativistic electrons and positrons emitting Cerenkov radiation.

The material properties of BGO are $X_0^{BGO} = 1.118 \text{ cm}$
with a density of $\rho = 7.13 \text{ g/cc}$

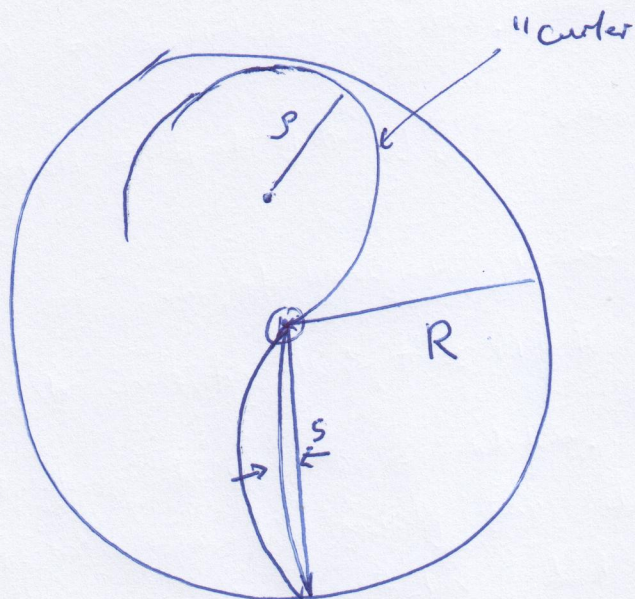
BGO = bismuth germanate



Lead glass. $X_0^{\text{Pb-glass}} = 1.265 \text{ cm}$
 $\rho = 6.22 \text{ g/cc}$

So Lead-glass and BGO are relatively similar in terms of compactness for EM showers.

BGO gives a light-yield of about 21% of NaI(Tl)
corresponding to about 11,500 photons per MeV of energy deposit.



$$R = 1.1 \text{ m}$$

$$B = 3.8 \text{ T}$$

i) Radius of curvature, ρ , of curving track, $\rho = \frac{R}{2} = 0.55 \text{ m}$

$$p_T = 0.2998 B R \rho$$

(with p_T in GeV/c , B in T, R in m)

$$= 0.2998 \times 3.8 \times 0.55$$

$$= 0.627 \text{ GeV}$$

ii) See notes from class 11.

~~$s = 2R \sin \frac{\theta}{2}$~~ For $L \ll \rho$. Here $L = R = 1.1 \text{ m}$.

$$p_T = 45 \text{ GeV}/c \Rightarrow \rho = \frac{p_T}{0.2998 B} = \frac{45}{0.2998 \times 3.8} = 39.5 \text{ m.}$$

So indeed, $L \ll \rho$,
and $s \approx \frac{L^2}{8\rho} = \frac{0.2998 B L^2}{8 p_T}$

So for $p_T = 45 \text{ GeV}/c$, $s = 3.83 \text{ mm}$

and for $p_T = 450 \text{ GeV}/c$ $s = 383 \mu\text{m}$

(12) continued.

So we basically measure the inverse P_T , often denoted, k

$$k \equiv \frac{1}{P_T} \text{ using the sagitta.}$$

$$\text{ie } k = \frac{1}{P_T} = \frac{8 \text{ s}}{0.2998 \text{ BL}^2} = 5.8035 \text{ s} \quad (\text{for our choice here of } B, L).$$

Now the sagitta is measured with ^{approximately} Gaussian uncertainties.

$$\text{ie } \Delta k = \Delta\left(\frac{1}{P_T}\right) = 5.8035 \Delta s$$

$$\text{But } k = \frac{1}{P_T}$$

$$\text{So } \Delta k = \frac{\Delta P_T}{P_T^2}$$

$$\text{ie } \Delta\left(\frac{1}{P_T}\right) = \frac{1}{P_T} \left(\frac{\Delta P_T}{P_T}\right)$$

$$\text{So } \left(\frac{\Delta P_T}{P_T}\right) = P_T \Delta\left(\frac{1}{P_T}\right) = P_T (5.8035 \Delta s).$$

Hence for a 1% error on $\frac{\Delta P_T}{P_T}$ for ~~$P_T = 45 \text{ GeV}/c$~~ ,

$$\text{So } \Delta s = \frac{0.01}{P_T} \times \frac{1}{5.8035}$$

$$\text{So for } P_T = 45 \text{ GeV}/c, \text{ Need } \Delta s = 38.3 \mu\text{m}$$

$$\text{and for } P_T = 450 \text{ GeV}/c, \text{ Need } \Delta s = 3.83 \mu\text{m}.$$

It gets very challenging to measure high momentum tracks with high precision. $\Rightarrow$ the ~~tracks~~ small track curvature is difficult to distinguish from a straight line.